Bruce ATS™ TotalViewITCH Specification

Revision [3.3]

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Overview

Bruce ATS TotalViewITCH is a direct data feed product offered by Bruce ATS providing the following features for all exchange-listed equities securities traded on the ATS:

- **Order Level Data:** Bruce ATS provides depth of book on an order-by-order basis using the standard ITCH format. TotalViewITCH uses a series of messages to track the life of a customer order, covering activities such as order entry, executions, replaces, and cancels.
- **Trade Messages:** To ensure that customers have complete information about Bruce ATS activity, TotalViewITCH supports a trade correction message to reflect adjustments to previously communicated executions.
- **Administrative Messages**:
    - *Stock Directory* - provide basic security data for the available symbols
    - *Trading Action* - inform market participants when a security is halted or released for trading
    - *Reg SHO Restriction* - provide the current RegSHO short sell restrictions being enforced
- **Event Controls:** Used to communicate key market events on Bruce ATS including start and end of day.

Architecture

The TotalViewITCH feed is made up of a series of sequenced messages. Each message is variable in length based on the message type. The messages that make up the TotalViewITCH protocol are typically delivered using a higher level protocol that takes care of sequencing and delivery guarantees.
Bruce ATS offers the TotalView ITCH data feed in two deliver protocol options:

- SoupBinTCP
- MoldUDP64

In the market data messages, instruments are identified by a stock locate code – a low lying integer employed with the intent of serving as an array index for rapidly looking up instrument details. The locate codes are dynamically assigned each day, starting with a value of 1, and communicated via the Stock Directory message. An instrument's locate code will not change intraday. However, there should be no expectation that the assignment will remain the same across multiple days. The Stock Locate code appears in all messages, and at the same position in all messages to support efficient filtering. A default value of 0 will be applied to messages which are not stock dependent.

Data Types

All integer fields are big-endian (network byte order) binary encoded numbers. They are unsigned, unless otherwise noted.

All alphanumeric fields are ASCII fields which are left justified and padded on the right with spaces.

Prices are integer fields, supplied with an associated precision. When converted to a decimal format, prices are in fixed point format, where the precision defines the number of decimal places. For example, a field flagged as Price (4) has an implied 4 decimal places.

Timestamps reflect the system time when the outbound message was generated and are represented as the number of nanoseconds since the start of the Unix epoch (January 1, 1970).

Message Formats

1.1 System Event Message

The System Event message is used to communicate key market or data feed handler events.

System Event Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"S"	System Event Message
Stock Locate	1	2	Integer	Always 0
Timestamp	3	8	Timestamp	Nanoseconds since Unix epoch
Event Code	11	1	Alphanumeric	See System Event Codes below

For each trading session, Bruce ATS will transmit the following event code messages:

Code	Explanation
"O"	*Start of Transmission*: Denotes that TotalViewITCH has started its daily transmission schedule
"S"	*Start of System Hours*: Denotes that Bruce ATS is open and ready to start accepting orders.
"Q"	*Start of Market Hours*: Denotes the start of Bruce ATS Market Hours and orders are now available for execution.
"M"	*End of Market Hours*: Denotes the end of Bruce ATS Market Hours and orders are no longer available for execution.
"E"	*End of System Hours*: Denotes that Bruce ATS is now closed and will not accept any new orders.
"C"	*End of Transmissions*: Denotes that TotalViewITCH has completed its daily transmission schedule.

Administrative Messages

1.1.1 Stock Directory

At the start of each trading session, Bruce ATS disseminates stock directory messages for all active symbols in its trading system.

Stock Directory

Name	Offset	Length	Type	Value/Description
Message Type	0	1	"R"	Stock Directory Message
Stock Locate	1	2	Integer	Locate Code uniquely assigned to the security symbol for the day.
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Stock	11	8	Alphanumeric	Denotes the security symbol for the issue in the Nasdaq execution system.
Market Category	19	1	Alphanumeric	Indicates Listing market or listing market tier for the issue

Code	Definition
A	NYSE ARCA
N	NYSE
P	**NYSE ARCA**
Q	NASDAQ
V	Investors' Exchange LLC
Z	CBOE BZX Exchange
<space>	Not available

Round Lot Size	20	4	Integer	Denotes the number of shares that represent a round lot for the issue
Authenticity	24	1	Alphanumeric	Flags whether an issue is treated by Bruce ATS as a live/production symbol or a test symbol

Code	Definition
P	Live/Production
T	Test

1.1.2 Stock Trading Action

Bruce ATS uses this administrative message to indicate the current trading status of a security to the trading community.

Prior to the start of system hours, Bruce ATS will send out a Trading Action spin in which TotalViewITCH will deliver a Stock Trading Action message with a *Trading State* of "T" for all securities that are eligible for trading at the start of the market hours. If a security is absent from the pre-opening Trading Action spin, firms should assume that the security is being treated as halted in the Bruce ATS platform.

Afterwards, Bruce ATS will use the Trading Action message to relay changes in trading status for an individual security. Messages will be sent when a stock is:

- Halted
- Released for trading

Stock Trading Action				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"H"	Stock Trading Action Message.
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Stock	11	8	Alphanumeric	Stock symbol, right padded with spaces
Trading State	19	1	Alphanumeric	Indicates the current trading state for the stock. Allowable values: "H" = Halted/Paused on Bruce ATS "T" = Trading on Nasdaq

1.1.3 Reg SHO Short Sale Price Test Restricted Indicator

In February 2011, the Securities and Exchange Commission (SEC) implemented changes to Rule 201 of the Regulation SHO (Reg SHO). For details, please refer to SEC Release Number 34-61595. In association with the Reg SHO rule change, Nasdaq will introduce the following Reg SHO Short Sale Price Test Restricted Indicator message format. For all issues, Bruce ATS relays the Reg SHO Short Sale Price Test Restricted Indicator message when it receives an update from the primary listing exchange.

Bruce ATS processes orders based on the most Reg SHO Restriction status value.

Reg SHO Restriction				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"Y"	Reg SHO Short Sale Price Test Restricted Indicator
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Stock	11	8	Alphanumeric	Stock symbol, right padded with spaces
Reg SHO Action	19	1	Alphanumeric	Denotes the Reg SHO Short Sale Price Test Restriction status for the issue at the time of the message dissemination. Allowable values are: "0" = No price test in place "1" = Reg SHO Short Sale Price Test Restriction in effect due to an intraday price drop in security "2" = Reg SHO Short Sale Price Test Restriction remains in effect

1.2 Add Order Message

An Add Order Message indicates that a new order has been accepted by the Nasdaq system and was added to the displayable book. The message includes a day-unique Order Reference Number used by Nasdaq to track the order. Nasdaq will support two variations of the Add Order message format.

1.2.1 Add Order

This message will be generated for orders accepted by the Bruce ATS system and added to the book.

Add Order Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"A"	Add Order
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Order Reference Number	11	8	Integer	The unique reference number assigned to the new order at the time of receipt.
Buy/Sell Indicator	19	1	Alphanumeric	The type of order being added. "B" = Buy Order. "S" = Sell Order.

Add Order Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Shares	20	4	Integer	The total number of shares associated with the order being added to the book.
Stock	24	8	Alphanumeric	Stock symbol, right padded with spaces
Price	32	8	Price (4)	The display price of the new order

1.3 Modify Order Messages

Modify Order messages always include the Order Reference Number of the Add Order to which the update applies. To determine the current display shares for an order, ITCH subscribers must deduct the number of shares stated in the Modify message from the original number of shares stated in the Add Order message with the same reference number. Nasdaq may send multiple Modify Order messages for the same order reference number and the effects are cumulative. When the number of display shares for an order reaches zero, the order is dead and should be removed from the book.

1.3.1 Order Executed Message

This message is sent whenever an order on the book is executed in whole or in part. It is possible to receive several Order Executed Messages for the same order reference number if that order is executed in several parts. The multiple Order Executed Messages on the same order are cumulative.

Order Executed Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"E"	Order Executed Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Order Reference Number	11	8	Integer	The reference number of the order being executed
Executed Shares	19	4	Integer	The number of shares executed
Match Number	23	8	Integer	The Bruce ATS generated day unique Match Number of this execution. The Match Number is also referenced in the Trade Break Message

1.3.2 Order Cancel Message

This message is sent whenever an order on the book is modified as a result of a partial cancellation.

Order Cancel Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"X"	Order Cancel Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)

Order Cancel Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Order Reference Number	11	8	Integer	The reference number of the order being partially canceled
Cancelled Shares	19	4	Integer	The number of shares being removed from the order

1.3.3 Order Delete Message

This message is sent whenever an order on the book is being cancelled. All remaining shares are no longer accessible so the order must be removed from the book.

Order Delete Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"D"	Order Delete Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Order Reference Number	11	8	Integer	The reference number of the order being deleted

Order Replace Message

This message is sent whenever an order on the book has been cancel-replaced. All remaining shares from the original order are no longer accessible and must be removed. The new order details are provided for the replacement, along with a new order reference number which will be used henceforth. Since the side, stock symbol cannot be changed by an Order Replace event, these fields are not included in the message. Firms should retain the side and stock symbol from the original Add Order message.

Order Replace Message				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"U"	Order Replace Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Original Order Reference Number	11	8	Integer	The original order reference number of the order being replaced
New Order Reference Number	19	8	Integer	The new reference number for this order at time of replacement Please note that the Nasdaq system will use this new order reference number for all subsequent updates
Shares	27	4	Integer	The new total displayed quantity
Price	31	8	Price (4)	The new display price for the order

1.4 Trade Messages

1.4.1 Trade Correction Message

In the rare case that an adjustment to previously communicated trade / order execution, the Trade Correction Message will be used to communicate the required updates.

Trade Correction Messages should be included in Bruce ATS time-and-sales displays as well as volume and other market statistics. Since Trade Correction Messages do not affect the book, however, they may be ignored by firms just looking to build and track the Nasdaq execution system display.

Trade Correction Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"C"	Trade Correction Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Match Number	11	8	Integer	The Bruce ATS match number of original execution
New Match Number	19	8	Integer	The Bruce ATS match number assigned to the adjusted execution
Shares	27	4	Integer	The shares associated with the adjusted execution
Price	31	8	Price (4)	The price associated with the adjusted execution

1.4.2 Trade Break Message

The Broken Trade Message is sent whenever an execution on Bruce ATS is broken. A trade break is final; once a trade is broken, it cannot be reinstated.

Firms that use the ITCH feed to create time---and---sales displays or calculate market statistics should be prepared to process the broken trade message. If a firm is only using the ITCH feed to build a book, however, it may ignore these messages as they have no impact on the current book.

Trade Break Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"B"	Trade Break Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp (see Data Types section for details)
Match Number	11	8	Integer	The Bruce ATS match number of the execution that was broken

Support

- For general product support and technical support for Nasdaq data feeds, please contact Nasdaq Global Information Services at clientsuccess@nasdaq.com

Appendices

Appendix A - Documentation Version Control Log

Version	Date Approved	Comments
1.0	2024-08-02	Initial Version
3.3	2026-08-11	Minor edits for clarity. Changed Nomenclature from BOSS to Bruce ATS in numerous places. The version number jump is to keep it in line with other market data specs.

Appendix B - Stock Symbol Convention

For all securities, Bruce ATS abides by any regulatory halts and trading pauses instituted by the primary or listing markets as appropriate. In addition, Bruce ATS may also halt trading for operational reasons. In all cases, Bruce ATS will send out a trading action message to inform its market participants when the trading status of an issue changes. For informational purposes, Bruce ATS also attempt to provide the reason for each trading action update, using a 4-byte reason code for this purpose.